An Information-Theoretic View of Cache-Aided Networks: Part 2 – Decentralized Coded Caching

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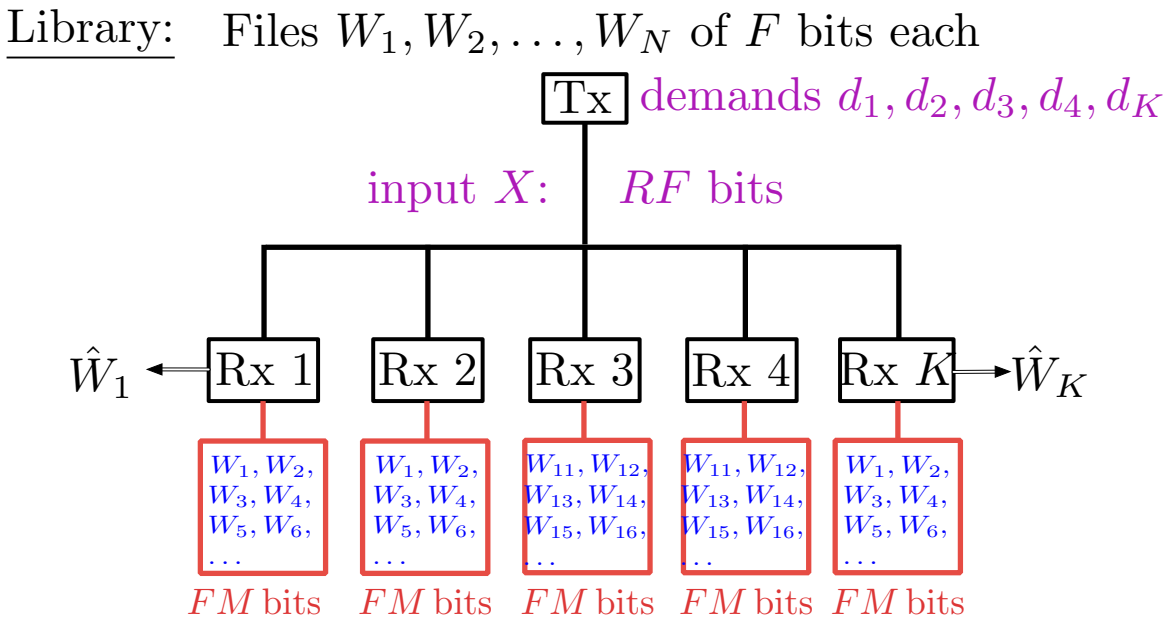
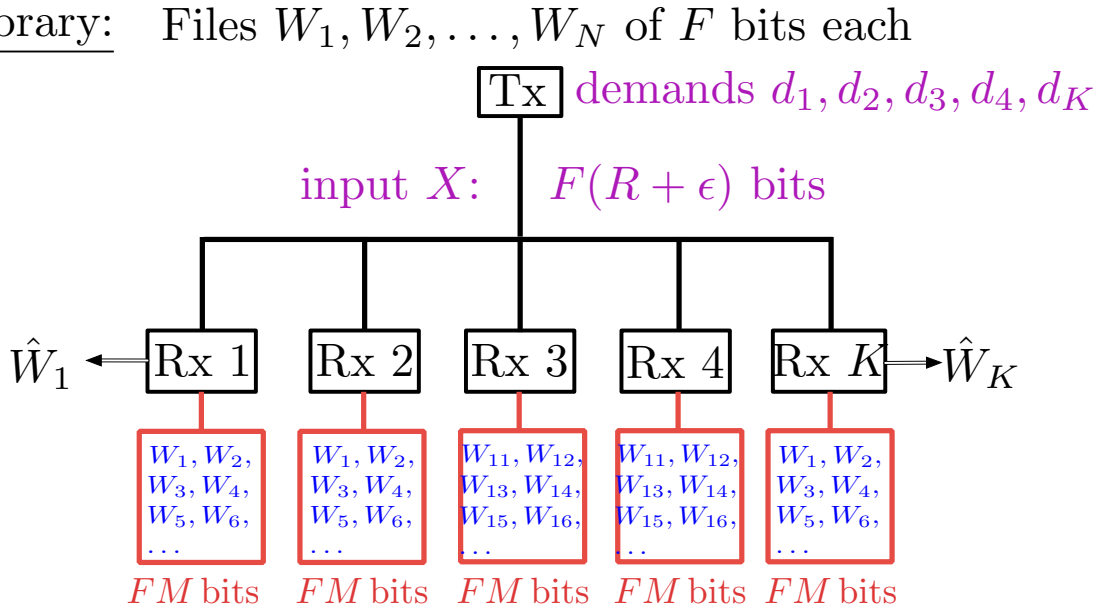
Decentralized Placement



Communication in two phases:

- Placement phase: Each receiver randomly downloads bits from library without knowing demands or the number of receivers K!
  $\rightarrow Z_k = g(W_1, \ldots, W_N, \Theta_k)$
- Delivery phase:
  - Receiver k wants file W_{d_k} $\rightarrow$ sends demand d_k to transmitter
  - Tx describes $W_{d_1}, \ldots, W_{d_K}$ to Rxs $1, \ldots, K$ through input X

Formal Problem Statement



- Cache placement $\quad Z_k = g(W_1, \ldots, W_N, \Theta_k)$,

  where Θ_k is a randomness known to everyone

- Delivery encoding $\quad X = f(W_1, \ldots, W_N, d_1, \ldots, d_K, \Theta_1, \ldots, \Theta_K)$

- Delivery decoding $\quad \hat{W}_k = \varphi_k(X, Z_k, d_1, \ldots, d_K, \Theta_1, \ldots, \Theta_K)$.

- Goal: $\quad \hat{W}_k = W_{d_k}$ for all $k = 1, \ldots, K$ $\quad$ with high probability

Fundamental Rate-Memory Tradeoff $R^\star_{\text{Dec}}(M)$

$$R^\star_{\text{Dec}}(M) \quad := \quad \min\Big\{R\colon \text{ such that for } (R,M) \text{ each receiver} k \in \{1,\dots,K\} \text{ learns } W_{d_k} \text{ with high probability } \Big\}$$

Obvious bounds:

- Local caching gain achievable $R^\star_{\text{Dec}}(M) \leq min\{K,N\}\left(1-\frac{M}{N}\right)$
- Cannot improve on centralized setup: $R^\star_{\text{Dec}}(M) \geq R^\star(M)$.

Decentralized Coded Caching Algorithm

- Placement: Each Receiver sequentially samples and stores each bit of the library with probability $p = \frac{M}{N}$

- For each subset $\mathcal{S} \subseteq \{1, \dots, K\}$, define now $W_d^{\mathcal{S}}$ the set of all bits of message W_d exclusively cached at all receivers of set $\mathcal{S}$.

- *Delivery Encoding:*
  - Send all demanded bits that are not cached anywhere

  - For $t = 1, \dots, K-1$ use the coded caching delivery scheme of parameter t to send the demanded bits

    $$\left\{ W_{d_1}^{\mathcal{S}}, \dots, W_{d_K}^{\mathcal{S}} : \qquad |\mathcal{S}| = t \right\}$$

    Zero-padding might be required for this operation!

  - Delivery decoding similar to coded caching scheme, but again for all parameters $t = 0, 1, 2, \dots, K-1$.

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[2] M. A. Maddah-Ali, U. Niesen, "Decentralized coded caching attains order-optimal memory-rate tradeoff," *IEEE Trans. on Inf. Theory*.

Analysis of Decentralized Scheme

- By the weak law of large numbers, for all $\epsilon > 0$:

$$\Pr\left[\left|\left|W_d^{\mathcal{S}}\right| - p^{|\mathcal{S}|}(1-p)^{K-|\mathcal{S}|}F\right| > \epsilon\right] \to 0 \quad \text{as} \quad F \to \infty.$$

- Expected storage

$$\begin{aligned} M &= N\sum_{t=1}^{K}\binom{K-1}{t-1}p^t(1-p)^{K-t} \\ &= Np\sum_{t'=0}^{K-1}\binom{K-1}{t'}p^{t'}(1-p)^{K-1-t'} = Np = M. \end{aligned}$$

- Expected rate:

$$\begin{aligned} R &= \sum_{t=0}^{K-1}\binom{K}{t+1}p^t(1-p)^{K-t} = \frac{1-p}{p}\sum_{t'=1}^{K}\binom{K}{t'}p^{t'}(1-p)^{K-t'} \\ &= \frac{1-p}{p}\sum_{t'=0}^{K}\binom{K}{t'}p^{t'}(1-p)^{K-t'} - \frac{1-p}{p}(1-p)^K \\ &= \frac{1-p}{p}(1-(1-p)^K) = \frac{N-M}{M}\left(1-\left(1-\frac{M}{N}\right)^K\right) \end{aligned}$$

Results for Decentralized Caching

Upper Bound $R^\star(M)$ for Decentralized Caching

$$R^\star(M) \leq K\left(1 - \frac{M}{N}\right) \min\left\{\frac{N}{KM}\left(1 - \left(1 - \frac{M}{N}\right)^K\right), \frac{N}{K}\right\}$$

- Above upper bound matches $R^\star(M)$ up to a factor of at least 12. (Proved analytically in [2].)
- Above upper bound matches coded caching upper bound for centralized scenario up to a factor of 1.6. (Shown numerically.)